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Studierichting: Informatica
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$$P(z) = z^5 + 3z^4 + 4z^3 + az^2 + bz + c$$

$$(z+1)^3 \mid P(z)$$

$$P(-1) = 0$$

$$P'(z) = 5z^4 + 12z^3 + 12z^2 + 2az + b$$

$$P'(-1) = 0$$

$$P''(z) = 20z^3 + 36z^2 + 24z + 2a$$

$$P''(-1) = 0$$

$$-2 + a - b + c = 0$$

$$5 - 2a + b = 0$$

$$-8 + 2a = 0$$

$$a = 4$$

$$b = 3$$

$$c = 1$$

References